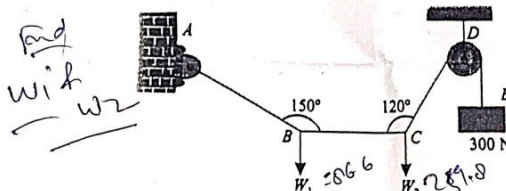
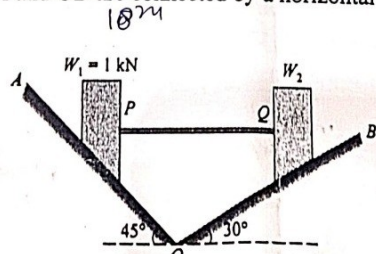


UNIT 2: Two-Dimensional Force Systems & Friction

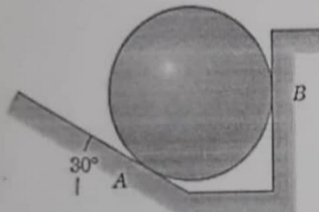
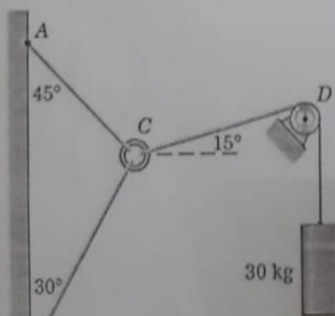
Syllabus topics: Laws of motion; principle of transmissibility; resultant of concurrent & non-concurrent force systems; distributed force system; free body diagrams; equilibrium; Lami's theorem, Varignon's theorem; laws of Coulomb friction; belt friction.

Question 1

1b	What are the three modes of heat transfer? Explain each briefly.	3	CO1	K2
2a	A light string ABCDE whose extremity A is fixed, has weights W_1 and W_2 attached to it at B and C. It passes round a small smooth peg at D carrying a weight of 300 N at the free end E as shown in Fig. 1.  Figure 1.	5	CO2	K3
2b	Two loads, W_1 (equal to 1 kN) and W_2, resting on two inclined rough planes OA and OB are connected by a horizontal link PQ as shown in Fig. 2. Two loads, W_1 (equal to 1 kN) and W_2, are resting on two inclined rough planes OA and OB are connected by a horizontal link PQ as shown in Fig. 2.  Figure 2.	5	CO2	K3

MID TERM-I | NME 102 | 2025-26 | Q2(a) String ABCDE with pulley, Q2(b) Two loads on inclined planes with link PQ

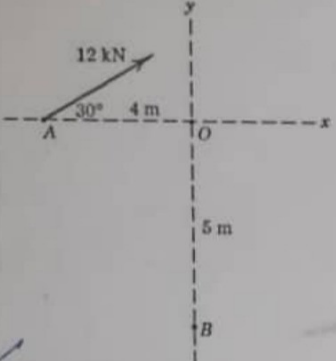
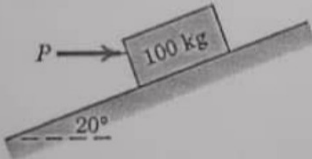
Question 2

		Harcourt Butler Technical University Kanpur		MID TERM-I		
Branch	All Technology (CHE, PL, BE, FT, PT, LFT, OT, BS-MS, BT)			Program	B. Tech	
Course Name	Introduction to Mechanical Engineering			Semester	1st	
Course Code	NME 101			Year	I	
Time: 1:00 Hr	Answer All Questions			Session	2025-26	
Knowledge Level (KL)	K1: Remembering		K3: Applying	K5: Evaluating		
	K2: Understanding		K4: Analysing	K6: Creating		
Maximum Marks:15, Attempt all Questions						
Q. No	Questions			Marks	COs	KL
1a	Find the magnitude of two equal forces acting at a point with an angle of 60° between them, if the resultant is equal to $30\sqrt{3}$ N.			2	CO2	K2, K4, K5
1b	State Varignon's Theorem.			1	CO2	K1
1c	The 50 kg homogeneous smooth sphere rests on the incline <i>A</i> and bears against the smooth vertical wall <i>B</i> . Draw the free body diagram of the sphere and calculate reactions at <i>A</i> and <i>B</i> .			2	CO2	K1, K2, K3
						
2a	Three cables are joined at the junction ring <i>C</i> . Determine the tensions in cables <i>AC</i> and <i>BC</i> caused by the weight of the 30-kg cylinder.			3	CO2	K3, K4, K5
						

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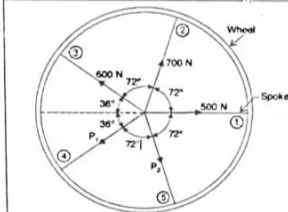
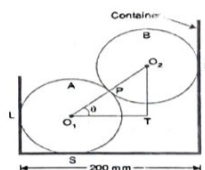
NME 101 | All Technology (CHE/PL/BE/FT/PT/LFT/OT/BS-MS/BT) | 2025-26 | Q1(a) Resultant of two forces, Q1(b) Varignon's theorem, Q1(c) Sphere on incline against wall, Q2(a) Three cables at ring C

Question 3

2b	Replace the 12-kN force acting at point A by a force-couple system at (a) point O and (b) point B. 	2	CO2	K2, K3, K5
3	Determine the magnitude and direction of the friction force acting on the 100 kg block shown if, first, $P=500$ N and, second, $P=100$ N. The coefficient of static friction is 0.20, and the coefficient of kinetic friction is 0.17. The forces are applied with the block initially at rest. 	5	CO2	K1, K2, K4, K6

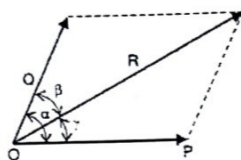
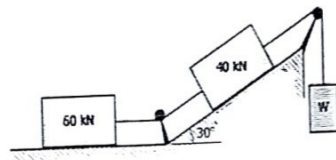
NME 101 | All Technology | 2025-26 | Q2(b) Force-couple system replacement, Q3 Friction force on 100kg block

Question 4

		Harcourt Butler Technical University Kanpur		1st MID SEM
Branch	(BE/BS-MS/BT/CH/FT/LT/OT/PL/PT)		Program	B. Tech
Course Name	INTRODUCTION TO MECHANICAL ENGINEERING		Semester	I
Course Code	NME-101		Year	1 ST
Time: 1:00 Hr	Maximum Marks: 15		Session	2024-25
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	
Note: Attempt any three questions. Each question carries the marks as shown against them.				
Q. No	Questions	Marks	COs	KL
1	Explain the following with neat sketches: a) Principle of Transmissibility b) Lami's Theorem	2 3	CO2	K1, K2
2	A wheel has five equally spaced radial spokes, all in tension. If the tensions of three consecutive spokes are 500 N, 700 N and 600 N respectively, find the tensions P_1 and P_2 in the other two spokes. 	5	CO2	K3, K5
3	A body of weight 300 N is pulled up along an inclined plane having inclination 60° to the horizontal. Find the force required to pull the body if the coefficient of friction is 0.3. If the distance travelled by body is 15 m along the plane, find the work done on the body.	5	CO2	K3, K5
4	Two spheres A and B weighing 100 N and 75 N respectively and with the corresponding radius 75 mm and 50 mm are placed in a container as shown in diagram. Determine the support reactions. 	5	CO2	K3, K5
Course Outcomes	CO1	Understand the basic laws concepts of mechanical systems.		
	CO2	Determine resultants and apply conditions of static equilibrium to plane force systems.		
	CO3	Analyze beam for shear force and bending moment along the span and analyze trusses for axial forces		
	CO4	Evaluate the structural properties centroid and moment of inertia		
	CO5	Stress analysis for one- and two-dimensional stress systems.		

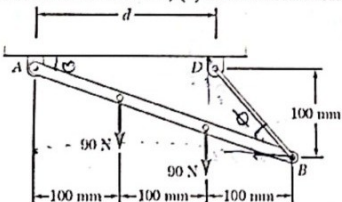
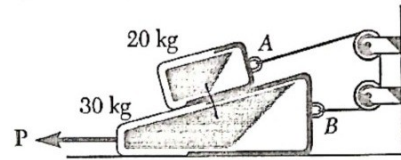
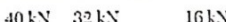
1st MID SEM | NME-101 | Technology | 2024-25 | Q1 Transmissibility & Lami's theorem, Q2 Wheel with radial spokes, Q3 Body pulled up incline, Q4 Two spheres in container

Question 5

		Harcourt Butler Technical University Kanpur		Ist MID SEM		
Branch		CHE, PT, BE, OT, LT, PL, BSMS		Program	B.Tech & BSMS	
Course Name		Introduction to Mechanical Engineering		Semester	I	
Course Code		NME 101		Year	First	
Time: 01.00Hr		Answer All Questions		Maximum Marks	15	
Note: 1. Marks of each question are shown against them. 2. Assume missing data if any.						
Q. No		Questions			Marks	COs
1		Discuss the laws of mechanics in brief? Two forces P and Q are acting at a point O as shown in Figure 1. The force P = 264.9 N and force Q = 195.2 N. If the resultant of the forces is equal to 400 N then find the values of angles β, γ, α.  Figure-1			5 [2+3]	CO2
2		What weight W is necessary to start the system of blocks shown in Figure-2 moving to the right? The coefficient of friction is 0.10 and the pulleys are assumed to be frictionless.  Figure-2			5	CO2

1st MID SEM | NME 101 | CHE/PT/BE/OT/LT/PL/BSMS | Q1 Two forces P & Q with resultant, Q2 Weight W for pulley system

Question 6

	determine (a) the tension in cable BD, (b) the reaction at A. 	5	CO	2 / 17
2b	The coefficients of static and kinetic friction are $\mu_s = 0.4$ and $\mu_k = 0.3$, respectively, between all surfaces of contact. Determine the force P for which motion of the 30 kg block is impending if cable AB (a) is attached as shown, (b) is removed. Assume that the cord is inextensible and there is no friction between the pulleys and the cord. 	5	CO2	K2, K3, K4
3a	For the beam and loading 			

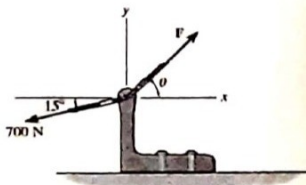
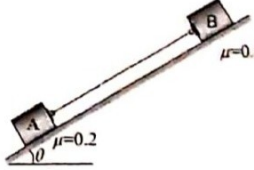
NME 101 | 2025-26 | Q2(a) Rod AB hinged with cable BD, Q2(b) Block on pulley system with friction

Question 7

1.a	Explain various modes of heat transfer with application.	5	CO1	K1, K2
1.b	Define different types of manufacturing techniques in brief.	5	CO1	K1, K2
2.a	Two identical rollers, each of weight 500 N, are supported by an inclined plane making an angle of 30° to the horizontal and a vertical wall as shown in the Figure. i) Sketch the free body diagrams of the two rollers. ii) Assuming smooth surfaces, find the reactions at the support points.	5	CO2	K3, K5
2.b	Discuss Coulomb's law of friction. A block of mass 10 kg is kept on a rough inclined plane as shown in figure. A force 3N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force for equilibrium? Draw shear force and bending moment diagram for following	5	CO2	K3, K5

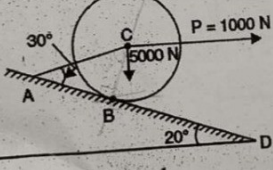
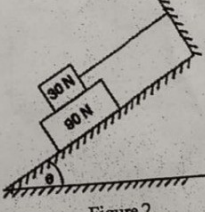
END SEM | NME-101 | Q2(a) Two identical rollers on incline & wall, Q2(b) Block on rough incline with friction

Question 8

3(a)	The plane as shown in Figure 1. The coefficient of friction between block A and the plane is 0.2, and that between block B and the plane is 0.5. Determine the value of the angle θ at which sliding will occur. Determine the tension in the rope.   Figure 1 Figure 2	5	CO3
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END SEM | NME-102 | 2022-23 | Q2(a) Resultant force along y-axis, Q2(b) Two blocks tied by rope with friction

Question 9

2	(a) A right circular roller of weight 5000 N rests on a smooth inclined plane and is held in position by a cord AC as shown in Figure 1. Find the tension in the cord if there is a horizontal force of magnitude 1000 N acting at C. (b) What should be the value of the angle θ in Figure 2 so that the motion of the 90 N block impends down the plane? The coefficient of friction μ for all the surfaces is $1/3$.   Figure 1. Figure 2.	5+5	CO2	K4, K3
	(a) A cantilever 1.5 m long is loaded with a uniformly distributed load of 2 kN/m. It also carries a point load of 1.25 m from the free end. Draw the shear force and bending moment diagram.	5+5	CO3	K5, K4

End Semester | Q2(a) Roller held by cord AC on incline, Q2(b) Angle θ for block motion with friction

